RoBuff

Pinch now, grip 2.56 seconds later.

Your bare hand, seen by a laptop webcam, drives a simulated robot arm on the Moon through the true Earth-Moon light delay, with a ghost arm that answers at once.

Ahmed Sohail Butt, solo. UB Hacking 2026.

The problem: light is slow at Moon distance

2.56 s

Earth to the Moon and back at the speed of light (mean distance 384,400 km, NASA NSSDC Moon fact sheet).

> In a 2024 University of Bristol study, a robot arm run at delays up to 2.6 s got worse on every measure the authors recorded.
>
> Louca et al., ACM Transactions on Human-Robot Interaction (abstract).

It is physics, not bandwidth: no network upgrade removes it.

RoBuff is designed for students and science-centre visitors to feel it with only a laptop webcam. Nobody has used it yet.

How it works: one loop, Sense, Understand, Act

Sense

Webcam frames go to the MediaPipe hand landmarker on your own machine: 21 points per hand.

Fake-camera test: median 11 ms per frame, a hand in 123 of 123 frames (RTX 4050).

→

Understand

Palm position and pinch. A guardrail verdict for any requested object. The ghost's error in millimetres.

→

Act

Inverse kinematics on MuJoCo's own Jacobian, position servos, then friction contact.

Physics at 240 frames per second headless (RTX 4050).

No camera frames leave the laptop. Gemini, when used, receives only the typed or spoken request text.

Hand control: common, and said so

Hand tracking is only the input. What it drives is new here: an arm that grips by friction, so it can slip.

- Pinch closes the jaw below 0.25 and opens it above 0.38, once 2 frames agree. Set from a sample photo, not yet tuned on a real pinch.
- Nothing glues the rock to the jaw: no weld, no parenting. A scripted pick of the 20 g cube rose 78.2 mm and held for 5.65 s.
- It is not robust off the centre line: 6 of 9 cube positions passed.
- Space stops the arm, and so do two open hands (experimental, `?hands=2`). Only Resume on screen clears a stop.

SIMULATION Measured with a scripted hand and a fake camera; the author has also steered it with his own hand on a webcam (not timed).

Delay and the ghost

- Moon mode: your commands wait 1.28 s to reach the arm, and the view waits 1.28 s to reach you.
- The ghost is a second simulation that runs your commands at once, drawn see-through over the delayed arm.
- It does not know the rock. The true rock has density 2.3 g/cm³ and friction 0.6; the ghost assumes 1.6 and 0.8. The gap between them is shown in millimetres.
- The literature says prediction helps reaching more than contact, so the grasp is where the ghost can be wrong.

> Not a new idea: DLR flew predictive graphics on the ROTEX arm in 1993 at a 5 to 7 s delay, after 1980s work at JPL and MIT. New here is a laptop webcam and a browser.

Objects on request, checked before they appear

Gemini proposes; code decides.

The receipt ticks each check against the simulated arm's measured limits (jaw 10 to 70 mm, up to 0.30 kg):

- ✓ Mesh is sane
- ✓ Convex hull
- ✓ Jaw direction
- ✓ Scale
- ✓ Mass
- ✓ Friction
- ✓ Grip height
- ✓ Simulated lift test

The lift test runs the same arm on a hidden copy of the world; only a rise of at least 48 mm at the end of the hold passes.

- "An asteroid the size of Texas": too wide as asked, rescaled to a 55 mm model of 144 g, which lifted 76.0 mm.
- "Solid iron, 60 mm": refused at 562 g.
- "A shard of comet ice": refused. The pads meet it 14 degrees off square, so it needs friction 0.26; ice has 0.03.

With a key, Gemini turns the words into a spec under a JSON schema. Without one, an offline keyword mock does, and the screen says which.

Gravity, changed live

Place	m/s^2	Source
Earth	9.82	NASA NSSDC
Mars	3.73	NASA NSSDC
Moon	1.62	NASA NSSDC
Bennu	8.3e-5	calculated from JPL data, spin ignored
Orbit	0	set

- A cube dropped from 0.10 m landed after 0.354 s on the Moon (theory 0.351 s) and 0.144 s on Earth.
- Bennu's escape speed is about 0.20 m/s. Flick the rock faster and the screen says it is gone. That is a label: the sim's gravity is uniform.
- Contact at very low gravity is not trustworthy: a 5 cm/s push lifted the resting cube 46.9 mm on Bennu. The screen warns below 0.1 m/s^2.

Results

Ghost gate: not yet measured.

0 logged trials, so I claim nothing about the ghost. The pass rule was written down before any data, and the result will be reported either way.

Measured so far, engineering checks only (not user results):

- Scripted friction pick: rose 78.2 mm, held 5.65 s.
- Lift test, Bennu-like sample: rose 77.3 mm, held 2.02 s, both jaws touching.
- Gravity drops match theory: 0.354 s against 0.351 s on the Moon.
- Hand tracking: median 11 ms per frame on a fake camera.

Disclosure and credits

- **Real:** your hand, read on your own machine.
- **Simulated:** both arms, the delay, the gravity and the contact. No real robot.
- **Mocked when no key is present:** object specs (offline keyword mock) and the voice (browser voice or cached lines). Labelled on screen.
- **New this weekend:** all application code, from an empty repo; first commit 1:43 PM on November 7, 2026.
- **Existed before:** the MuJoCo Menagerie SO-ARM100 model (Apache-2.0), MuJoCo 4.0.0, MediaPipe, three.js, and NASA and JPL data.
- **My earlier, different projects:** WorkCell, EmberHacks, HandSignal and FleetLine. No code from them is reused.
- **AI tools:** Claude Code planned, summarised research, and wrote code, tests and docs under my direction and review. Gemini works inside the product on object specs. MediaPipe reads the hand.
- **Credits:** every library, model, data source and licence is in THIRD_PARTY.md.
- **Code:** github.com/ahmedsohail2003/buffalo-hackathon-2